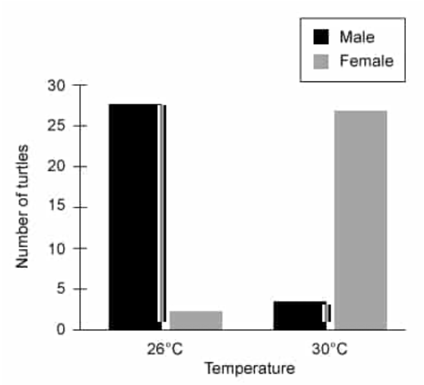
Certain turtle species exhibit embryonic expression of an enzyme that converts male sex hormones to female sex hormones. The enzyme expression level determines whether a developing embryo will be male or female. Scientists hypothesize that expression of this enzyme depends on incubation temperature during development. To test this, they incubated 30 turtle eggs at 26°C and 30 eggs at 30°C and recorded the sex of each turtle after hatching (shown below).

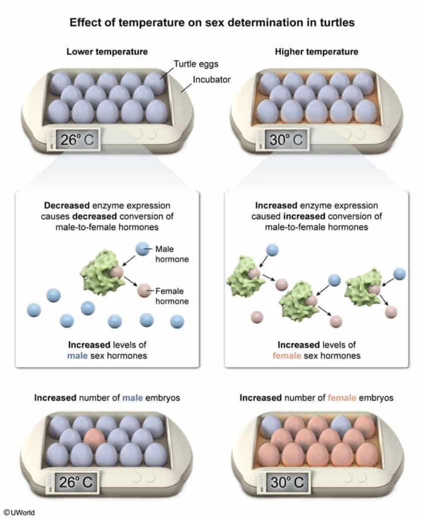


Given this, the expression level of this enzyme in developing turtles is most likely:

- A. high at 26°C and low at 30°C. [4%]
- ✔ B. low at 26°C and high at 30°C. [88%]
- C. high at both 26°C and 30°C. [6%]
- D. low at both 26°C and 30°C.

Correct
88% answered correctly

Explanation:



An organism's genome comprises the complete set of genes possessed by that organism, and the expression of these genes produces proteins that confer various traits (ie, phenotypes) to the organism. However, certain environmental conditions (eg, temperature, diet, oxygen level) may influence gene expression or alter protein function, affecting an organism's phenotype. This relationship between an organism's genetic make-up and the environment is known as a **gene-environment interaction**.

In this scenario, certain turtles experience temperature-dependent sex determination. This is an example of a gene-environment interaction during a developmental process. *Unlike humans*, the sex of the developing embryos for these turtles is determined by the temperature at which the egg is incubated, *not* by the presence or absence of specific sex chromosomes.

The turtles express **an enzyme** during embryonic development that **converts male sex hormones** (ie, testosterone) **to female sex hormones** (ie, estrogen). This enzyme's expression level (and corresponding amounts of male or female sex hormones) determines whether an embryo develops as male or female. Accordingly, *high* expression of this enzyme in embryos *increases* the conversion of male to female sex hormones, increasing the presence of female hormones and inducing *female* development. In contrast, *low* expression of this enzyme *decreases* the conversion of male to female sex hormones, increasing the presence of male hormones and inducing *male* development.

The scientists hypothesized that expression of this enzyme in turtle embryos depended on incubation temperature. To test this, they incubated groups of turtle eggs at either 26°C or 30°C and recorded the sex of each turtle after hatching. The graph shows that *most* embryos in eggs **incubated at 26°C developed as male**, signifying that **enzyme levels** would be **low at 26°C** because *reduced* expression of this enzyme promotes male development. In contrast, the graph shows that *most* embryos in eggs **incubated at 30°C developed as female**, meaning that **enzyme levels** would be **high at 30°C** because *elevated* enzyme levels promote female development **(Choices A, C, and D)**.

Educational objective:
In general, the expression of genes produces proteins, and the properties and activities of these proteins lead to the expression of certain traits (phenotypes) in organisms. However, in a gene-environment interaction, certain environmental conditions (eg, temperature, diet, oxygen level) can influence gene expression or protein activity and alter an organism's phenotype.

Time spent:0 sec
Subject:
Biology

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System:
Reproduction